MALIGNANT FIBROUS DERMAL TUMORS

AT A GLANCE

A group of uncommon and rare cutaneous tumors with varying degrees of malignant potential.

Includes **dermatofibrosarcoma protuberans**, **atypical fibroxanthoma**, **malignant fibrous histiocytoma**, **fibrosarcoma**, and **epithelioid sarcoma**.

Diagnosis is made by biopsy.

- **Dermatofibrosarcoma protuberans (DFSP)** is a locally aggressive tumor with a high rate of local recurrence and rare metastases.
- **Atypical fibroxanthoma** is an intermediate-grade neoplasm that usually arises on sun-damaged skin in the eighth decade.
- **Malignant fibrous histiocytoma** represents a morphologic pattern shared by many poorly differentiated neoplasms.
- **Epithelioid sarcoma** is a rare tumor that classically presents on the hands and fingers of young males and can mimic nonneoplastic inflammatory lesions.

Etiology and Pathogenesis

The histogenesis of DFSP is uncertain: fibrohistiocytic, purely fibroblastic, and neural-related differentiation have all been hypothesized. A very sensitive, although non-specific, marker for DFSP is **CD34**. A distinctive cell population within normal nerves has been shown to express CD34, consistent with a possible neural-related histogenesis of DFSP. CD34 can help distinguish between DFSP and scar tissue in re-excision specimens.

At the molecular level, DFSP exhibits the reciprocal translocation **t(17;22)(q22;q13)**, producing a fusion of the *collagen type I a1* gene and the *platelet-derived growth factor B-chain* (*PDGFB*) gene. Current techniques permit sensitive and specific detection of these fusion transcripts in paraffin-embedded DFSP tumor specimens by reverse transcription polymerase chain reaction (RT-PCR). Both fibrosarcomatous transformation of DFSP and some cases of superficial fibrosarcoma harbor these fusion transcripts, suggesting a close affinity between DFSP and fibrosarcoma.

Clinical Findings

History

DFSP is a slow-growing lesion that often presents on the trunk and proximal extremities. Due to its indolent onset, patients may present for evaluation when the tumor is several centimeters in size. The tumor is often misdiagnosed as a simple scar or cyst. It may grow to several centimeters in diameter and form large, infiltrative tumors. In a small subset of patients, there is a history of prior trauma at the site. DFSP has also been reported to arise in scars, including surgical, burn, and vaccination sites.

Cutaneous Lesions

The clinical morphology of DFSP is variable. The most common presentation is a firm, indurated plaque, often skin-colored, with red-brown exophytic nodules (see Fig. 125-1A). It can also present as a nonprotuberant, atrophic, violaceous lesion resembling morphea, sclerosing basal cell carcinoma, anetoderma, or scar (see Fig. 125-2). Large nodular tumors may develop. The epidermis may be atrophic, and focal ulceration can occur. The most common site is the trunk, followed by the extremities. Approximately 15% of cases occur on the head and neck.

Laboratory Tests

Histopathology: DFSP is diagnosed by skin biopsy.
Microscopically, DFSP is characterized by fibroblastic spindle cells arranged in

interlacing fascicles that intersect and bend at acute angles, producing a **storiform (starburst) pattern** (see Fig. 125-1B).